



بسم الله الرحمن الرحيم

Al Bayan University

Faculty Of Engineering

Telecommunication Department



GRADUATION PROJECT

TRAFFIC PREDICTION AND RESOURCE ALLOCATION IN 4G CELLULAR NETWORKS

USING LSTM

- ❖ **A Graduation Project Submitted to the Department of Electronic Systems and Communication Engineering
in Partial Fulfillment of the Requirements for the Degree of Bachelor of Science (B.Sc.)
in Communication Engineering**
-

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الإهداء

بسم الله الرحمن الرحيم

الحمد لله رب العالمين الذي وفقنا وأعاننا ومنَّ علينا بإتمام هذا العمل، والصلاة والسلام على سيدنا محمد ﷺ خير معلمٍ
وهادٍ إلى سبيل الرشاد.

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أن يحفظهما ويجزيهما عنا خير الجزاء.

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وامتنانًا.

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والحمد لله رب العالمين حمداً يوافي نعمه ويكافئ مزيده، والصلاة والسلام على معلم البشرية هادياً ومبشراً ونذيراً.
انطلاقاً من قوله صلى الله عليه وسلم: "من لا يشكر الناس لا يشكر الله"، نتقدم ببالح الشكر وعظيم الامتنان إلى صرحنا الأكاديمي الشامخ:

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Abstract

The rapid growth of mobile data traffic has created significant challenges in managing network resources efficiently in Fourth Generation (4G) cellular networks. Traditional resource allocation methods often rely on reactive strategies, which may lead to network congestion, increased latency, and inefficient utilization of available bandwidth. To address these challenges, this project proposes an intelligent traffic prediction approach based on the Long Short-Term Memory (LSTM) deep learning model.

The proposed system utilizes historical network traffic data to predict future traffic patterns and support proactive resource allocation decisions. The dataset is preprocessed through data cleaning, normalization, and sequence generation before being used to train and evaluate the LSTM model. The predicted traffic values can assist network operators in allocating radio resources more efficiently, reducing congestion, and improving the overall Quality of Service (QoS).

The performance of the proposed model is evaluated using standard regression metrics, including Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE). Experimental results demonstrate that the LSTM-based approach effectively captures temporal traffic patterns and provides accurate predictions, making it suitable for intelligent network resource management in modern wireless communication systems.

The proposed framework contributes to improving network efficiency, reducing resource wastage, and supporting the development of intelligent traffic management techniques for current and future mobile communication networks.

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List of Abbreviations

LTE	Long Term Evolution
4G	Fourth Generation Mobile Communication System
LSTM	Long Short-Term Memory
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
RNN	Recurrent Neural Network
CNN	Convolutional Neural Network
KPI	Key Performance Indicator
QoS	Quality of Service
SINR	Signal-to-Interference-plus-Noise Ratio
RSSI	Received Signal Strength Indicator
RSRP	Reference Signal Received Power
RSRQ	Reference Signal Received Quality
PRB	Physical Resource Block
CQI	Channel Quality Indicator
MSE	Mean Squared Error
RMSE	Root Mean Square Error
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
R ²	Coefficient of Determination
CPU	Central Processing Unit
GPU	Graphics Processing Unit

RAM	Random Access Memory
OS	Operating System
GUI	Graphical User Interface
Python	Python Programming Language
NumPy	Numerical Python Library
Pandas	Python Data Analysis Library
Matplotlib	Mathematical Plotting Library
Scikit-Learn	Machine Learning Library for Python
TensorFlow	Open-Source Deep Learning Framework
Keras	High-Level Neural Network API
Epoch	One Complete Training Cycle
Batch Size	Number of Training Samples Processed in One Iteration
Time Series	Sequence of Data Points Indexed in Time Order

Chapter One

1.1 Introduction

With the exponential growth of mobile devices, smart technologies, and high-bandwidth applications, modern cellular networks are facing unprecedented data traffic demands. Fourth-Generation Long-Term Evolution (4G LTE) networks remain a critical backbone of global telecommunications, handling a massive share of mobile data. However, the dynamic and irregular nature of mobile user behavior, spatial-temporal variations, and sudden spikes in network usage present significant challenges for network operators. Traditional static network management approaches often lead to either over-provisioning—which wastes expensive network resources or under-provisioning, resulting in network congestion, dropped calls, and poor Quality of Service (QoS).

To bridge the gap between network capacity and user demand, proactive network automation has become an absolute necessity. Cellular traffic prediction stands at the forefront of this shift, enabling operators to foresee traffic patterns and dynamically allocate resources before bottlenecks occur. By accurately forecasting future traffic loads, mobile operators can implement efficient network optimization strategies such as dynamic base station sleep scheduling, adaptive bandwidth allocation, and intelligent load balancing. This not only guarantees a seamless user experience but also significantly reduces operational expenditures (OPEX) and energy consumption.

In recent years, deep learning has emerged as a powerful paradigm for modeling complex and non-linear patterns in time-series data. Traditional statistical methods such as ARIMA often fail to capture the long-term dependencies and highly volatile fluctuations inherent in 4G LTE traffic. To overcome these limitations, Long Short-Term Memory (LSTM) networks, which are a specialized form of Recurrent Neural Networks (RNNs), have been widely adopted. LSTMs are capable of learning sequential dependencies and preserving historical information over long periods, making them highly effective for forecasting daily, weekly, and seasonal mobile traffic trends.

This research focuses on developing an intelligent framework for 4G LTE traffic prediction and network optimization using LSTM networks. By leveraging historical traffic datasets, the proposed model predicts future traffic loads with high accuracy. Based on these predictions, a proactive optimization mechanism dynamically adjusts network configurations to improve efficiency and service quality.

Consequently, there is a growing need for intelligent prediction models capable of learning complex temporal traffic patterns and providing accurate forecasts. Deep learning techniques, particularly Long Short-Term Memory (LSTM) networks, have emerged as powerful solutions for addressing these challenges due to their ability to model sequential data and capture long-term dependencies. This research investigates the application of LSTM networks for intelligent traffic prediction and proactive resource allocation in 4G cellular networks.

The primary contributions of this research are summarized as follows:

- Analyzing real-world 4G LTE traffic data to identify spatial and temporal patterns.
- Developing and fine-tuning an LSTM-based deep learning model for accurate multi-step traffic forecasting.
- Proposing a proactive optimization strategy to improve resource utilization and minimize network congestion.

1.2 Problem Statement

The core research problem arises from several major challenges in modern 4G LTE network management, including:

1.2.1 Inadequacy of Traditional Models

Traditional statistical approaches such as ARIMA are unable to effectively capture the highly dynamic, non-linear, and temporal-spatial characteristics of modern cellular traffic

1.2.2 Resource Allocation Dilemma

Static resource allocation strategies often lead to:

Over-provisioning during low-traffic periods, causing unnecessary energy and financial waste.

Under-provisioning during peak hours, resulting in congestion, dropped calls, and degraded user experience.

1.2.3 Reactive Network Management

Most current optimization techniques respond to congestion only after it occurs, which limits the ability to maintain real-time service quality

1.2.4 Lack of Integrated Intelligent Frameworks

There is a shortage of automated systems capable of integrating deep-learning-based traffic forecasting with proactive network optimization and real-time resource management

1.3 Research Objectives

1.3.1 Main Objective

The primary objective of this research is to develop a proactive, data-driven framework capable of accurately forecasting 4G LTE traffic and dynamically optimizing network resources .using deep learning techniques

1.3.2 Specific Objectives

The specific objectives of this research are:

To analyze and preprocess real-world 4G LTE traffic datasets in order to identify spatial-temporal patterns and anomalies.

To design and implement an LSTM-based prediction model capable of accurate multi-step-ahead forecasting.

To evaluate the performance of the proposed model using metrics such as RMSE, MAE, and R^2 .

To compare the proposed LSTM model with traditional statistical methods such as ARIMA.

To develop a proactive optimization mechanism based on traffic predictions for efficient resource allocation.

To reduce operational costs and energy consumption while improving Quality of Experience (QoE) for users.

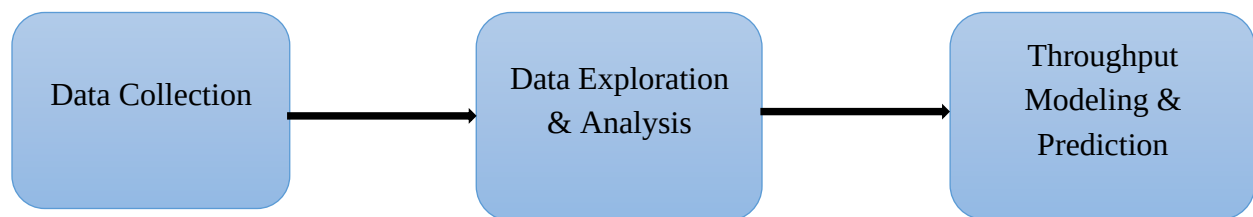


FIGURE 1 : RESEARCH FRAMEWORK STRUCTURE.

- **Development of an Intelligent Traffic Prediction Framework:** A deep learning framework based on Long Short-Term Memory (LSTM) networks is proposed to accurately predict future traffic patterns in 4G cellular networks.

- **Comprehensive Data Preprocessing Pipeline:** A systematic preprocessing workflow was designed, including data cleaning, normalization, feature engineering, and sequence generation, to improve the quality and consistency of the training dataset.
- **Design of an Optimized Deep LSTM Architecture:** A stacked LSTM model enhanced with Batch Normalization, Dropout, and Dense layers was developed to effectively capture temporal dependencies while reducing overfitting.
- **Comprehensive Performance Evaluation:** The proposed model was evaluated using multiple regression metrics, including MAE, MSE, RMSE, MAPE, and the coefficient of determination (R^2), providing a reliable assessment of prediction accuracy.
- **Support for Intelligent Resource Allocation:** The predicted traffic information can be utilized to enable proactive resource allocation, helping reduce network congestion and improve Quality of Service (QoS).
- **Scalable Framework for Future Networks:** Although the framework is developed for 4G cellular networks, its architecture can be extended and adapted to emerging 5G and future 6G communication systems.

1.5 Significance of the Research

The importance of this research lies in its contribution to transforming network management from a reactive process into an intelligent and proactive system. The significance can be summarized in the following points:

1.5.1 Enabling Intelligent Network Automation

This study demonstrates how deep learning models such as LSTM can be integrated into real telecom systems for automated resource management.

1.5.2 Reducing Operational Costs and Energy Consumption

Accurate traffic prediction enables operators to reduce unnecessary energy usage through mechanisms such as dynamic base station sleep scheduling, thereby lowering OPEX and supporting green communication initiatives.

1.5.3 Improving Quality of Service (QoS) and Quality of Experience (QoE)

The proactive nature of the proposed framework helps prevent congestion before it occurs, ensuring stable connectivity and higher user satisfaction.

1.5.4 Supporting Future Network Evolution

Although the study focuses on 4G LTE networks, the developed predictive and optimization framework can be extended to future 5G and 6G heterogeneous network environments.

1.6 Research Methodology

This research adopts a structured, data-driven methodology implemented using Python programming and advanced deep learning libraries. The methodology consists of the following phases:

1.6.1 Data Preparation and Preprocessing

Importing and handling LTE traffic datasets using Pandas and NumPy.

Handling missing values using interpolation techniques.

Extracting temporal features such as:

Hour of the day

Day of the week

Weekend indicators

Applying normalization using MinMaxScaler

Splitting the dataset into:

Training set (70%)

Validation set (15%)

Testing set (15%)

1.6.2 LSTM Model Architecture Design

Building the deep learning model using TensorFlow and Keras.

Reshaping data into a 3D tensor format: [Samples, Time Steps, Features]

Stacking multiple LSTM layers to capture long-term dependencies.

Applying Dropout layers to reduce overfitting.

Using a Dense layer for final prediction output.

Compiling the model using:

Adam Optimizer

Mean Squared Error (MSE) Loss Function

1.6.3 Performance Evaluation

The trained model is evaluated using unseen testing data through the following metrics:

Root Mean Squared Error (RMSE)

Mean Absolute Error (MAE)

Coefficient of Determination (R^2)

These metrics are calculated using Scikit-Learn evaluation modules in Python.

1.6.4 Proactive Optimization Engine

A predictive optimization mechanism is implemented using Python logic based on the LSTM outputs:

- **Dynamic Sleep Scheduling**

If predicted traffic drops below a predefined threshold, selected base stations are switched into low-power sleep modes.

- **Congestion Avoidance**

If future traffic spikes are predicted, the system proactively activates bandwidth expansion and load balancing mechanisms before congestion occurs.

1.7 Thesis Organization

This thesis is organized into six chapters to provide a logical and systematic presentation of the research.

- **Chapter One** introduces the research background, problem statement, objectives, scope, contributions, and thesis organization.
- **Chapter Two** reviews the theoretical background of 4G cellular networks, traffic prediction techniques, machine learning, deep learning, Long Short-Term Memory (LSTM) networks, and relevant previous studies.
- **Chapter Three** describes the research methodology, including dataset preparation, preprocessing, feature engineering, model architecture, training strategy, and evaluation metrics.
- **Chapter Four** presents the experimental setup, including the software environment, hardware specifications, model configuration, training procedure, and implementation details.
- **Chapter Five** discusses the experimental results, performance evaluation, prediction accuracy, error analysis, and resource allocation improvements achieved by the proposed framework.
- **Chapter Six** concludes the research by summarizing the findings, presenting the major contributions, discussing limitations, and suggesting future research directions.

Chapter Two

Theoretical Background and Literature Review

2.1 Introduction

This chapter presents the theoretical background related to cellular traffic prediction and network optimization. It focuses on LTE networks, Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL), and Long Short-Term Memory (LSTM) networks. In addition, the chapter reviews previous studies, experimental implementations, advantages, limitations, and modern applications of LSTM-based traffic prediction models.

2.2 Basic Concepts

2.2.1 Cellular Traffic Concept

Cellular traffic refers to the amount of data, voice calls, and messages transmitted through mobile communication networks during a specific period. Understanding traffic patterns is essential for improving network performance and ensuring efficient resource allocation.

2.2.2 Traffic Prediction Concept

Traffic prediction is the process of using historical network data to estimate future traffic demands. Accurate prediction helps network operators optimize resources and prevent congestion.

2.2.3 Network Optimization Concept

Network optimization refers to a set of techniques and strategies used to enhance network performance, increase efficiency, and improve Quality of Service (QoS).

2.3 Definitions

2.3.1 Long-Term Evolution (LTE)

Long-Term Evolution (LTE) is a fourth-generation (4G) wireless communication technology designed to provide high-speed internet access, low latency, and improved network capacity.

2.3.2 Artificial Intelligence (AI)

Artificial Intelligence is a branch of computer science that enables machines and systems to simulate human intelligence and perform tasks such as decision-making and problem-solving.

2.3.3 Machine Learning (ML)

Machine Learning is a subset of AI that allows systems to learn from data and improve their performance without being explicitly programmed.

2.3.3.1 Supervised Learning

Supervised learning is the task of learning a function that maps the input features to desired output values. Supervised learning algorithms receive a labeled dataset, where each sample in the dataset has a corresponding label or ground truth. Supervised learning can be further divided into classification and regression:

- **Classification:** Classification models approximate a mapping from the input variables to a discrete output variable. A classification model classifies the inputs into one of two or more classes. The performance of the model is often measured by the classification accuracy.

- **Regression:** Regression models approximate a mapping from the input variables to a continuous output variable. The performance of a regression model is measured in terms of errors made in the model's predictions.

2.3.3.2 Unsupervised Learning

Unsupervised learning algorithms receive a set of input variables with no output variable or label. The objective of unsupervised learning is to learn the underlying structure of the data and find an efficient representation for it. Two common unsupervised learning tasks are clustering and dimensionality reduction:

- **Clustering:** Clustering groups samples into clusters based on their similarities [19]. Samples which are similar to each other are grouped into the same cluster.
- **Dimensionality Reduction:** The idea of dimensionality reduction is to project samples from a high-dimensional space onto a lower dimensional space without losing much information [20]. This reduces the complexity of the data while retaining its structure.

2.3.4 Deep Learning (DL)

Deep Learning is a specialized field of machine learning that uses multi-layer neural networks to learn complex patterns from large datasets.

2.3.5 Recurrent Neural Networks (RNN)

Recurrent Neural Networks are neural networks specifically designed for sequential and time-series data processing.

2.3.6 Long Short-Term Memory (LSTM)

LSTM is an advanced type of recurrent neural network developed to overcome the vanishing gradient problem and capture long-term dependencies in sequential data.

2.4 Historical Background

2.4.1 Evolution of Cellular Networks

First Generation (1G)

The first generation of cellular communication emerged in the 1980s and relied on analog communication technology.

Second Generation (2G)

The second generation introduced digital communication and enabled services such as text messaging.

Third Generation (3G)

The third-generation improved data transmission rates and supported mobile internet services.

Fourth Generation (4G LTE)

LTE technology was introduced around 2008 and significantly enhanced internet speed, network capacity, and user experience.

Fifth Generation (5G)

The fifth generation provides ultra-high-speed communication, low latency, and support for smart applications and the Internet of Things (IoT).

2.4.2 Evolution of Artificial Intelligence and Deep Learning

Artificial Intelligence research began in the 1950s. However, deep learning experienced rapid growth after 2010 due to advances in computational power, cloud computing, and the availability of large-scale datasets.

2.5 Types of Traffic Prediction Models

2.5.1 Statistical Models

Traditional statistical approaches include:

ARIMA (Auto Regressive Integrated Moving Average)

SARIMA (Seasonal ARIMA)

Holt-Winters Exponential Smoothing

These models are effective for linear time-series forecasting but may struggle with complex nonlinear patterns.

2.5.2 Machine Learning Models

Common machine learning techniques include:

Decision Trees

Random Forest

Support Vector Machines (SVM)

These methods can capture more complex relationships than traditional statistical models.

2.5.3 Deep Learning Models

Advanced deep learning models include:

Artificial Neural Networks (ANN)

Convolutional Neural Networks (CNN)

Recurrent Neural Networks (RNN)

Long Short-Term Memory Networks (LSTM)

Gated Recurrent Units (GRU)

These models are particularly effective for large-scale and highly dynamic datasets.

2.6 Long Short-Term Memory (LSTM) Networks and Their Operation

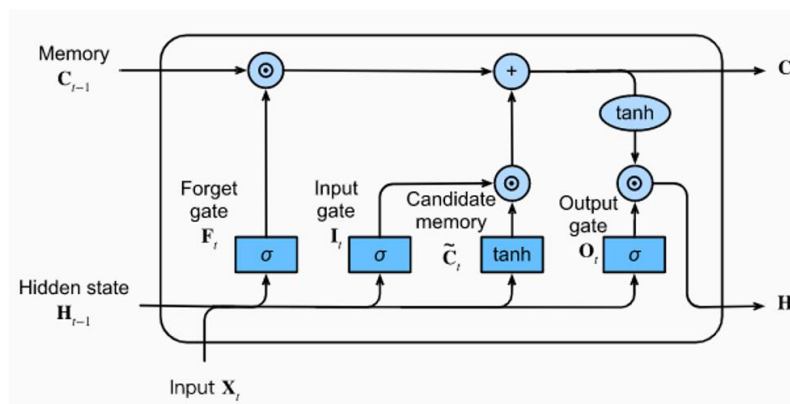


FIGURE 2 : LSTM STRUCTURE

2.6.1 Components of an LSTM Network

- **Forget Gate**
The forget gate determines which information should be removed from the cell state.
- **Input Gate**
The input gate controls the amount of new information added to the memory cell.
- **Cell State**
The cell state serves as the long-term memory of the network, preserving important information across time steps.
- **Output Gate**
The output gate determines what information will be passed to the next hidden state and the final output.

2.6.2 Working Mechanism of LSTM

The operation of an LSTM network generally involves the following steps:

- Receiving sequential input data.
- Evaluating information through the forget, input, and output gates.
- Updating the cell state.
- Producing future predictions based on learned temporal patterns.

2.7 Justification for Selecting LSTM

Although several machine learning and statistical techniques have been proposed for network traffic prediction, selecting an appropriate prediction model depends on the characteristics of the traffic data. Network traffic exhibits nonlinear behavior, temporal dependencies, and dynamic fluctuations that cannot be effectively captured by traditional statistical models.

Long Short-Term Memory (LSTM) networks were selected in this research because they are specifically designed to process sequential data and overcome the limitations of conventional Recurrent Neural Networks (RNNs), particularly the vanishing gradient problem. The memory cell mechanism enables LSTM models to retain important historical information over long periods, making them highly suitable for time-series forecasting.

Furthermore, previous studies have demonstrated that LSTM-based models consistently outperform many traditional prediction methods in terms of forecasting accuracy and robustness when applied to communication network traffic. Therefore, LSTM represents an

appropriate choice for developing an intelligent traffic prediction framework capable of supporting proactive resource allocation in 4G cellular networks.

2.8 Previous Experiments and Evaluations

2.8.1 Experiments Using Real LTE Traffic Datasets

Numerous studies have employed real-world LTE traffic datasets to evaluate prediction models. Results consistently show that LSTM-based models achieve lower prediction errors and higher forecasting accuracy than traditional methods.

2.8.2 Performance Evaluation Metrics

Several metrics are commonly used to assess prediction accuracy:

- Mean Absolute Error (MAE)
- Measures the average magnitude of prediction errors.
- Mean Squared Error (MSE)
- Calculates the average squared difference between predicted and actual values.
- Root Mean Squared Error (RMSE)
- Represents the square root of MSE and is widely used for model evaluation.
- Coefficient of Determination (R^2)
- Measures how well the model explains the variability in the observed data.

2.9 Introduction Previous Studies

A comprehensive review of previous studies is essential for understanding the current state of research in network traffic prediction. The reviewed literature highlights different approaches ranging from traditional statistical models to modern deep learning techniques. Analyzing these studies helps identify existing limitations and motivates the development of the proposed LSTM-based framework presented in this research.

Study	Authors	Year	Method	Dataset	Key Findings	Limitations / Research Gap
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Study 1	Kang, D., Lv, Y., Chen, Y.-Y., & Wang, F.-Y.	2017	LSTM Recurrent Neural Network	Traffic flow data	LSTM outperformed traditional models in short-term traffic prediction	Focused only on traffic flow, not cellular network resource allocation
Study 2	Trinh, H. D., Giupponi, L., & Dini, P.	2019	LSTM Networks	Raw mobile traffic data	Achieved high accuracy (94%) using raw data without manual feature extraction	Did not address resource allocation or network optimization
Study 3	Azzouni, A., & Pujolle, G.	2018	LSTM / GRU	Network traffic matrix	LSTM performed better than GRU for long-term predictions	Higher computational cost; limited to traffic matrix prediction only
Study 4	Ndolane, D., Massa, N., Dialo, D., et al.	2021	Machine Learning (Random Forest, SVM, etc.)	4G Orange Senegal dataset (100M records)	Identified CQI_Avg as the most influential feature; achieved 90.49% accuracy	Limited to feature importance analysis; did not include deep learning
Study 5	Li, A.-M., & Li, J.	2025	Hybrid LSTM + Transformer (HSLTNet)	Cellular network traffic	Achieved state-of-the-art spatiotemporal prediction	Complex architecture; requires high computational resources

Study 6	Hontinfinde, R. D., Tognisse, I. S., Tonou, M. M. S., et al.	2025	LSTM / BiLSTM + Dynamic Resource Allocation	Call Detail Records (CDR)	Achieved 81% BBU reduction and 88.34% resource utilization efficiency	Focused on C-RAN; limited to 4G networks
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TABLE 1 SUMMARY AND COMPARISON OF PREVIOUS STUDIES ON NETWORK TRAFFIC PREDICTION

2.9.1 Detailed Explanations of Each Study

1. Kang et al. (2017) – LSTM for Traffic Flow Prediction

Short-term Traffic Flow Prediction with LSTM Recurrent Neural Network

- **Contribution:** First comprehensive application of LSTM for short-term traffic flow prediction
- **Method:** LSTM with stacked architecture
- **Result:** Demonstrated superior performance over ARIMA and SVM
- **Limitation:** Focused only on urban traffic, not cellular networks

2. Trinh et al. (2019) – Mobile Traffic Prediction from Raw Data

Mobile Traffic Prediction from Raw Data Using LSTM Networks

- **Contribution:** Direct prediction from raw mobile network data
- **Method:** Deep LSTM architecture with minimal preprocessing
- **Result:** 94% prediction accuracy
- **Limitation:** No integration with resource allocation systems

3. Azzouni & Pujolle (2018) – LSTM for Network Traffic Matrix

A Long Short-Term Memory Recurrent Neural Network Framework for Network Traffic Matrix Prediction

- **Contribution:** Comprehensive framework for predicting traffic matrices
- **Method:** LSTM and GRU comparison
- **Result:** LSTM outperformed GRU for long-term predictions
- **Limitation:** High computational cost; limited scope

4. Ndolane et al. (2021) – 4G Network Feature Analysis

Finding Hidden Links among Variables in a Large-Scale 4G Mobile Traffic Network Dataset Using Machine Learning

- **Contribution:** Comprehensive feature importance analysis using 100M records
- **Method:** Random Forest, Decision Trees, SVM, Logistic Regression
- **Result:** CQI_Avg identified as most influential feature
- **Limitation:** Used traditional ML only; no deep learning application

5. Li & Li (2025) – Hybrid LSTM-Transformer Network

A hybrid spatiotemporal LSTM and transformer network for cellular traffic prediction

- **Contribution:** Novel hybrid architecture for spatiotemporal prediction
- **Method:** HSLTNet (LSTM + Transformer)
- **Result:** State-of-the-art performance
- **Limitation:** Complex architecture; high resource requirements

6. Hontinfinde et al. (2025) – Dynamic Resource Allocation

Dynamic resource allocation in cloud-radio access network using call detail record analysis

- **Contribution:** Integrated prediction with dynamic resource allocation
- **Method:** LSTM/BiLSTM + optimization algorithm
- **Result:** 81% BBU reduction; 88.34% utilization efficiency
- **Limitation:** Focused on C-RAN architecture

2.9 Research Gap

Based on the comprehensive review presented in Table 1 , several research gaps have been identified in the field of network traffic prediction and resource allocation.

First, most existing studies focus exclusively on improving prediction accuracy without considering how the predicted traffic information can be utilized to support intelligent resource allocation and network optimization. Studies such as those by Kang et al. (2017), Trinh et al. (2019), and Azzouni & Pujolle (2018) demonstrate high prediction accuracy but do not extend their frameworks to resource management.

Second, traditional machine learning approaches, as applied by Ndolane et al. (2021), fail to capture the complex temporal dependencies inherent in network traffic data, limiting their practical effectiveness.

Third, while recent hybrid architectures (Li & Li, 2025) achieve state-of-the-art performance, they introduce additional computational complexity that may hinder real-time deployment.

Fourth, some studies address network optimization (Cunha, 2024; Hontinfinde et al., 2025) but focus on energy saving or specific architectures such as C-RAN, rather than general 4G cellular network traffic prediction.

Finally, there is a lack of comprehensive evaluation using a diverse set of performance metrics (MAE, MSE, RMSE, MAPE, R^2) in many previous studies.

To address these limitations, the proposed work develops a complete framework that integrates:

1. Advanced LSTM deep learning for accurate traffic prediction
2. Comprehensive preprocessing and feature engineering for data quality
3. Multiple performance evaluation metrics for reliable assessment

4. Proactive resource allocation based on predicted traffic demand

This integrated approach ensures that the proposed framework is not only accurate but also practically applicable in real-world network management scenarios.

2.10 Advantages of LSTM-Based Traffic Prediction

- High prediction accuracy.
- Ability to capture temporal dependencies.
- Effective learning of complex traffic patterns.
- Improved network resource allocation.
- Reduced network congestion.
- Enhanced Quality of Service (QoS).
- Support for intelligent and autonomous networks.

2.11 Limitations and Challenges

Requirement for large amounts of training data.

- High computational cost.
- Long training times.
- Complexity in hyperparameter tuning.
- Challenges in real-time deployment.

2.12 Modern Applications

- LTE and 5G Networks
- Traffic load prediction.
- Radio resource management.
- Network fault detection.
- Congestion prevention.
- Intelligent Networks
- Self-Organizing Networks (SON).
- AI-driven network management systems.
- Internet of Things (IoT)
- Connected device management.

- Energy consumption optimization.
- Smart city infrastructure support.

2.13 Future Trends

Future research directions include:

- 6G network traffic forecasting.
- Federated Learning.
- Explainable Artificial Intelligence (XAI).
- Edge Computing.
- Digital Twin Networks.

2.14 Chapter Summary

This chapter reviewed the theoretical foundations of 4G cellular networks, network traffic prediction techniques, machine learning, deep learning, recurrent neural networks, and Long Short-Term Memory (LSTM) networks. In addition, a comprehensive review of previous studies was presented, followed by a comparative analysis highlighting existing research gaps.

Based on the findings of the literature review, the proposed LSTM-based framework was identified as a suitable solution for addressing the limitations of previous approaches. The next chapter describes the research methodology and implementation process adopted to develop the proposed prediction model.

Chapter Three

Research Methodology

This chapter presents the methodology adopted to develop the proposed intelligent traffic prediction and network optimization system for Fourth Generation (4G) cellular networks. The methodology describes the complete workflow of the study, beginning with dataset acquisition and preprocessing, followed by feature engineering, Long Short-Term Memory (LSTM) model development, model training, performance evaluation, and finally the utilization of predicted traffic values for network resource optimization.

The proposed framework aims to accurately forecast future network traffic using historical traffic observations and network performance indicators. Accurate traffic prediction enables proactive resource allocation, congestion reduction, improved Quality of Service (QoS), and more efficient utilization of available network resources. Figure 3.1 illustrates the overall workflow of the proposed methodology.

3.1 Research Methodology

This research follows a data-driven experimental methodology based on deep learning techniques for time-series forecasting. The proposed framework consists of several sequential stages designed to ensure reliable traffic prediction and effective resource optimization.

The methodology begins with collecting and preparing the network traffic dataset. The collected data are then preprocessed to remove inconsistencies, normalize numerical values, and encode categorical variables where required. After preprocessing, the data are transformed into sequential input samples suitable for training an LSTM neural network.

The developed LSTM model is trained using historical traffic observations to learn temporal traffic patterns and predict future network load. The trained model is subsequently evaluated using multiple statistical performance metrics, including Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and the Coefficient of Determination (R^2).

Finally, the predicted traffic values are incorporated into a resource allocation framework to support proactive bandwidth allocation, reduce network congestion, and improve overall network efficiency.

3.2 Proposed System Framework

The proposed system consists of six major stages that work together to perform intelligent traffic prediction and network optimization. The overall workflow is illustrated in Figure 3.1 and can be summarized as follows:

1. Dataset Collection
2. Data Preprocessing
3. Feature Engineering
4. LSTM Model Development
5. Traffic Prediction
6. Network Resource Optimization

Each stage contributes to improving the prediction accuracy and supporting intelligent decision-making for resource management in 4G cellular networks.

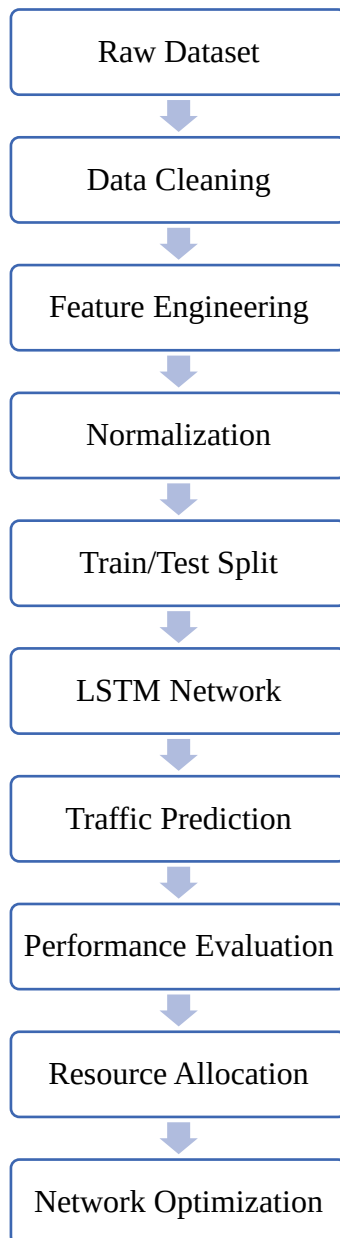


FIGURE 3: SYSTEM FRAMEWORK

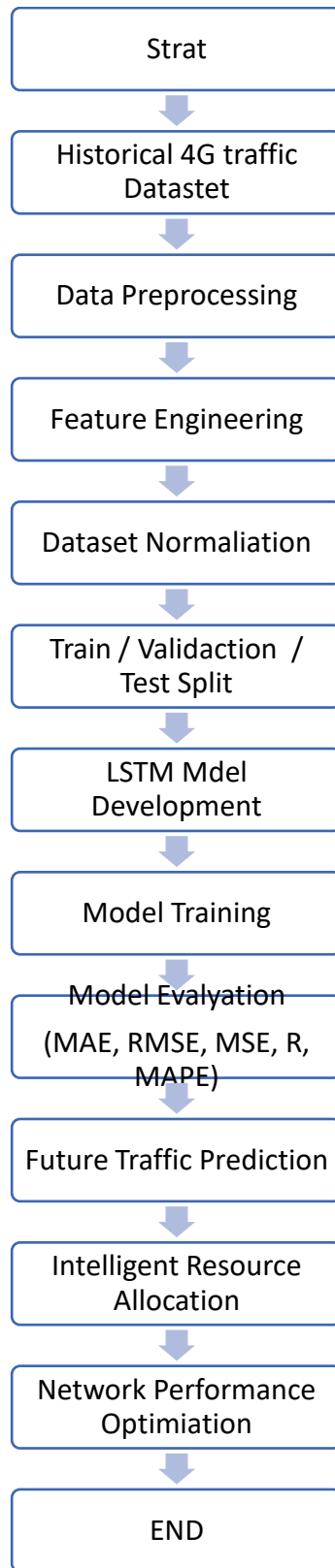


FIGURE 4: OVERALL RESEARCH METHODOLOGY

3.3 Data Description

The effectiveness of any deep learning model depends largely on the quality and diversity of the dataset used for training and evaluation. In this study, a comprehensive dataset representing the operational behavior of a Fourth Generation (4G) cellular network was employed to develop and evaluate the proposed traffic prediction framework.

The dataset contains 10,000 network observations, where each observation corresponds to a specific network state recorded at 15-minute intervals.

This temporal resolution allows the model to capture both short-term fluctuations and long-term traffic trends, making it suitable for time-series forecasting using the Long Short-Term Memory (LSTM) neural network.

Each observation consists of 18 features that describe different aspects of network operation. These features include temporal information, user activity, network utilization, traffic measurements, radio signal quality, Quality of Service (QoS) indicators, and resource allocation parameters. The diversity of these variables enables the model to learn complex relationships between traffic behavior and network performance.

The dataset contains both numerical and categorical variables. Numerical attributes describe measurable network characteristics such as traffic load, throughput, latency, packet loss, signal quality, and resource allocation, whereas categorical attributes provide contextual information including cell type, season, and weekend indicators. This combination improves the model's ability to recognize different traffic patterns under varying network conditions.

Feature	Description
Timestamp	Date and time of data collection
Hour	Hour of the day (0–23)
Day of Week	Day of the week
Is Weekend	Weekend indicator
Week of Month	Week number within the month
Season	Seasonal category
Cell Type	Cellular site classification

Feature	Description
Cell Capacity (MB)	Maximum available capacity
Traffic Load (MB)	Actual network traffic load
User Activity	Number of active users
Network Utilization (%)	Percentage of utilized resources
Throughput (Mbps)	Data transmission rate
Latency (ms)	End-to-end communication delay
Packet Loss (%)	Packet loss ratio
SINR (dB)	Signal-to-Interference-plus-Noise Ratio
Congestion Flag	Congestion status
Traditional Resource Allocation (MB)	Resource allocation using conventional techniques
Optimal Resource Allocation (MB)	Optimized resource allocation target

TABLE 2 . DATASET FEATURES

Property	Description
Dataset Type	Network Traffic Dataset
Network Generation	4G Cellular Network
Number of Samples	10,000
Number of Features	18
Target Variable	Traffic Load
Data Type	Time-Series
Missing Values	Removed During Preprocessing
Normalization	Min-Max Scaling
Training Set	70%
Validation Set	15%
Testing Set	15%

TABLE 3 . DATASET SUMMARY

Table 3 summarizes the main characteristics of the dataset used in this research. The dataset consists of time-series traffic observations collected from a 4G cellular network environment. After preprocessing, the dataset was divided into training, validation, and testing subsets to ensure reliable model development and unbiased performance evaluation.

3.4 Data Processing

Raw network traffic data often contain inconsistencies, redundant information, and variations in scale that may negatively affect the learning capability of deep learning models. Therefore, a comprehensive data preprocessing stage was performed before training the proposed LSTM model. The primary objective of this stage was to improve data quality, reduce noise, and transform the raw dataset into a structured format suitable for time-series prediction

3.4.1 Data Cleaning

The first step focused on improving the quality and consistency of the collected dataset. The dataset was examined to identify duplicated records, missing values, inconsistent measurements, and invalid entries that could introduce bias during model training.

The cleaning process included:

- Removing duplicate observations.
- Detecting and handling missing values.
- Eliminating corrupted or inconsistent records.
- Verifying data integrity and consistency.
- Ensuring chronological ordering of the observations.

These operations reduced data redundancy and ensured that the LSTM model learned meaningful traffic patterns rather than artifacts caused by poor-quality data

3.4.2 Missing Value Handling

Missing values frequently occur in communication network datasets because of temporary equipment failures, transmission interruptions, or incomplete measurements. Since missing observations can significantly degrade prediction performance, appropriate handling techniques were applied before model training.

Depending on the characteristics of the affected feature, one or more of the following techniques were considered:

- Mean value replacement.
- Linear interpolation.
- Forward filling.
- Backward filling.

Applying these techniques improved dataset completeness while preserving the temporal continuity required by the LSTM model

3.4.3 Categorical Feature Encoding

Several variables in the dataset, including (*Cell Type, Season, and Weekend Indicator*), represent categorical information that cannot be processed directly by neural networks.

These variables were transformed into numerical representations using suitable encoding techniques, allowing the LSTM model to utilize both numerical and categorical information during training.

The encoding process ensured that all input variables were represented in a machine-readable format while preserving their underlying relationships

3.4.4 Data Normalization

The numerical variables in the dataset have different measurement units and value ranges. For example, traffic load is measured in megabytes (MB), throughput in megabits per second (Mbps), latency in milliseconds (ms), and SINR in decibels (dB). Such differences may slow the training process and reduce prediction accuracy.

To overcome this issue, feature normalization was performed using the Min-Max Scaling technique, which transforms all numerical features into a common range between 0 and 1.

Normalization provides several advantages:

- Faster convergence during training.
- Improved numerical stability.
- Reduced gradient instability.
- Better prediction accuracy.
- More efficient learning process.

After normalization, all numerical variables contributed equally during model optimization.

3.4.5 Sequence Generation

Unlike conventional machine learning algorithms, LSTM networks require sequential input data to capture temporal dependencies.

Therefore, the normalized traffic observations were converted into overlapping time sequences. Each input sequence consisted of several consecutive historical observations, while the corresponding target represented the traffic value at the next prediction interval.

This transformation enabled the model to learn both short-term and long-term temporal relationships within the network traffic

3.4.6 Dataset Splitting

After preprocessing, the dataset was divided into three subsets to ensure reliable model evaluation.

- Training Set (70%) – Used for learning traffic patterns and updating model parameters.
- Validation Set (15%) – Used to monitor model performance during training and reduce overfitting.
- Testing Set (15%) – Used to evaluate the final predictive capability of the trained model using previously unseen data.

This splitting strategy ensures that the evaluation results accurately reflect the model's ability to generalize to new network conditions.

3.4.7 Preprocessing Workflow Summary

The preprocessing stage transformed raw network traffic measurements into structured sequential data suitable for LSTM-based time-series forecasting. Through cleaning, encoding, normalization, sequence generation, and dataset partitioning, the quality of the input data was significantly improved, enabling the proposed prediction model to achieve reliable and accurate forecasting performance.

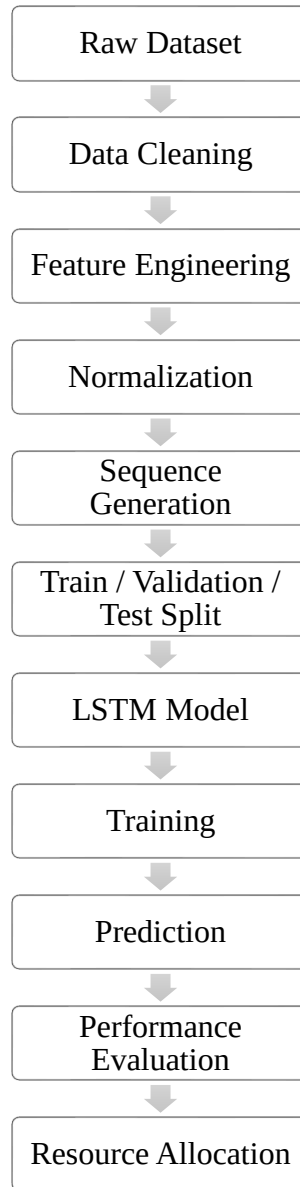


FIGURE 5 . WORKFLOW SUMMARY

Figure 5 illustrates the complete workflow adopted in this research, beginning with data preprocessing and ending with prediction-based resource allocation. Each stage contributes to improving the accuracy and reliability of the proposed framework

3.5 Feature Engineering

Feature engineering is a crucial stage in developing high-performance deep learning models. The objective of this stage is to prepare and organize the most informative variables that

enable the LSTM model to learn temporal traffic behavior effectively. Instead of using raw measurements directly, the selected features provide meaningful information that reflects user activity, network conditions, and traffic dynamics over time.

The proposed traffic prediction model utilizes both temporal and network-related features to improve prediction accuracy. Temporal features describe when traffic occurs, while network performance features describe the operational status of the cellular network.

The selected features are categorized as follows.

3.5.1 Temporal Features

Temporal variables help the model recognize periodic traffic behavior that naturally occurs during different hours, weekdays, weekends, and seasons. These variables include:

- Hour of the Day
- Day of the Week
- Week of the Month
- Weekend Indicator
- Season

These features allow the model to capture recurring traffic patterns such as daily peak hours and weekly usage trends.

3.5.2 Traffic Features

Traffic-related variables represent the current state of network demand and user activity.

The selected traffic features include:

- Traffic Load (MB)
- User Activity
- Network Utilization (%)

These variables directly describe the amount of traffic being handled by the network and are considered the primary indicators used by the LSTM model for future traffic prediction.

3.5.3 Quality of Service Features

Quality of Service (QoS) indicators reflect the operational quality experienced by users.

The selected QoS parameters include:

- Throughput (Mbps)
- Latency (ms)
- Packet Loss (%)
- SINR (dB)

These features provide valuable information regarding network efficiency and communication quality under different traffic conditions.

3.5.4 Resource Allocation Features

The dataset also contains information related to resource allocation strategies.

These variables include:

- Traditional Resource Allocation
- Optimal Resource Allocation

These features enable the proposed framework not only to forecast future traffic but also to support intelligent resource management decisions based on predicted demand.

3.5.5 Feature Selection Rationale

The selected features were chosen because they collectively represent the major factors influencing traffic behavior in 4G cellular networks.

Combining temporal information with QoS indicators and traffic statistics enables the LSTM model to capture both short-term fluctuations and long-term traffic dependencies.

Moreover, correlation analysis performed during the exploratory data analysis indicated that several variables, particularly Traffic Load, Network Utilization, User Activity, and Throughput, exhibit strong relationships with future traffic demand. This observation supports their inclusion in the prediction model and enhances the model's forecasting capability

Category	Features
Temporal Features	Hour, Day of Week, Week of Month, Season, Weekend Indicator
Traffic Features	Traffic Load, User Activity, Network Utilization
QoS Features	Throughput, Latency, Packet Loss, SINR
Resource Features	Traditional Allocation, Optimal Allocation

TABLE 4 SELECTED INPUT FEATURES

3.6 Exploratory Data Analysis (EDA)

Before developing the prediction model, an Exploratory Data Analysis (EDA) was conducted to understand the characteristics of the collected dataset, identify hidden patterns, detect anomalies, and examine the relationships among network performance variables.

EDA plays an essential role in deep learning applications because it provides valuable insights into data quality and feature behavior before model training. The analysis also supports feature selection and helps identify variables that significantly influence future traffic demand.

Several statistical and graphical visualization techniques were employed during this stage, including traffic trend analysis, network utilization distribution, correlation analysis, Quality of Service (QoS) evaluation, heatmap visualization, and statistical distribution analysis.

The following subsections summarize the most significant findings obtained from the exploratory analysis.

3.6.1 Traffic Trend Analysis

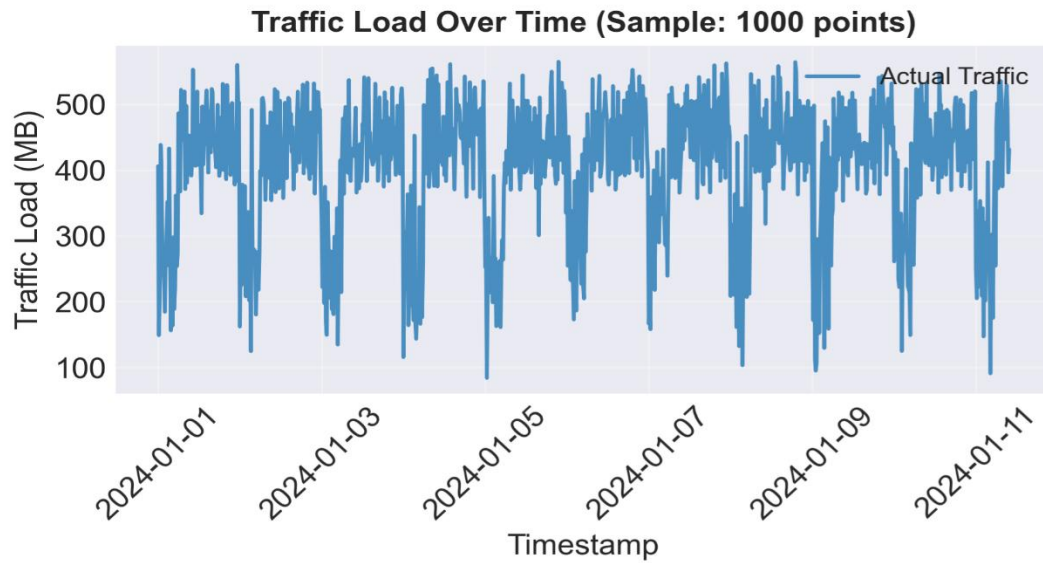


FIGURE 6 . TRAFFIC LOAD ANALYSIS

Figure 3.7 illustrates the variation in network traffic over the observation period. The traffic load exhibits clear temporal fluctuations with several peak periods corresponding to increased user activity. These recurring traffic patterns indicate that the dataset possesses strong temporal characteristics, making it highly suitable for sequence learning using the LSTM architecture.

The observed traffic dynamics further justify the use of deep learning techniques capable of capturing both short-term variations and long-term dependencies within sequential network data.

3.6.2 Network Utilization Analysis

4G Network Traffic Analysis Dashboard

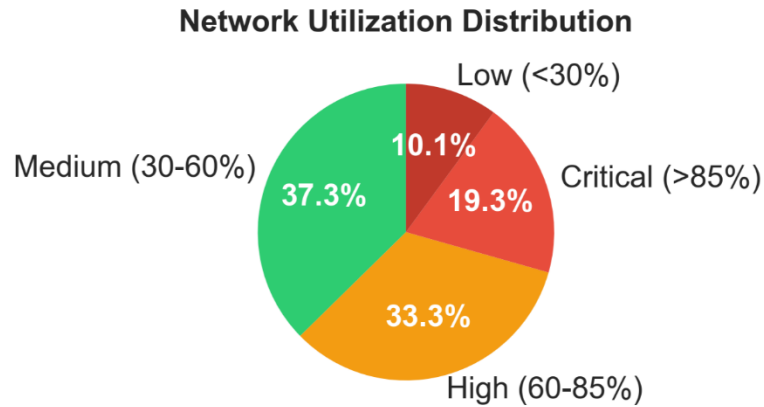


FIGURE 7 . NETWORK TRAFFIC ANALYSIS

The network utilization distribution reveals that most observations correspond to medium and high utilization levels, while relatively few samples belong to extremely low or overloaded conditions.

This balanced distribution provides sufficient representation of different network operating conditions, enabling the proposed prediction model to generalize effectively across varying traffic scenarios.

3.6.3 Correlation Analysis

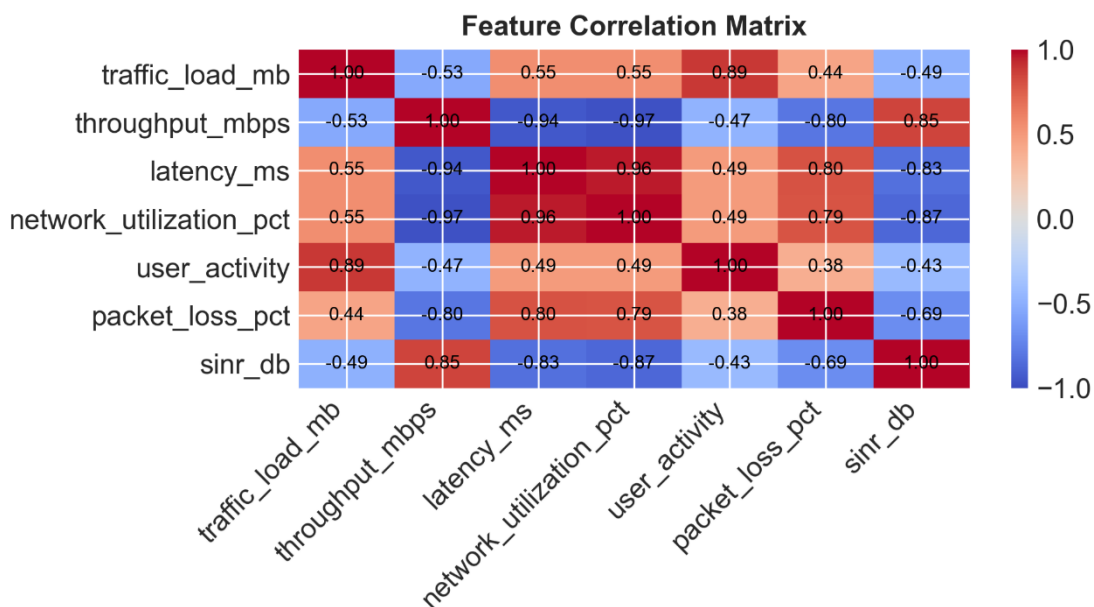


FIGURE 8 . CORRELATION ANALYSIS

The correlation matrix provides valuable insights into the relationships among the network variables. Strong positive correlations can be observed between Traffic Load, Network Utilization, and User Activity, indicating that increased user demand directly influences network resource consumption.

Conversely, certain Quality of Service parameters exhibit weaker or inverse correlations, highlighting their sensitivity to traffic congestion and varying network conditions.

These observations support the selection of the most informative features used for training the LSTM model.

3.6.4 QoS Analysis

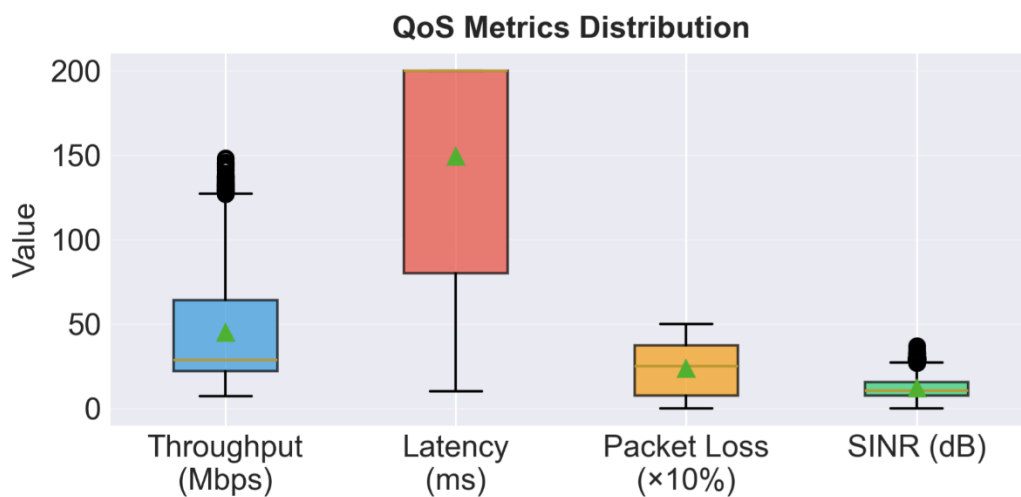


FIGURE 9 . QUALITY OF SERVICE ANALYSIS

The Quality-of-Service analysis demonstrates the statistical behavior of key network performance indicators, including throughput, latency, packet loss, and signal quality.

Most observations remain within acceptable operating ranges, while a limited number of outliers correspond to periods of heavy traffic congestion.

These results indicate that the dataset realistically represents practical network conditions encountered in operational cellular systems.

3.7 Proposed LSTM Architecture

The core component of the proposed intelligent traffic prediction framework is the Long Short-Term Memory (LSTM) neural network. LSTM is a specialized type of Recurrent Neural Network (RNN) designed to learn temporal dependencies in sequential data while overcoming the vanishing gradient problem commonly encountered in conventional RNN models.

Unlike traditional machine learning algorithms, LSTM networks maintain an internal memory that enables them to retain relevant historical information over long time intervals. This capability makes LSTM particularly suitable for network traffic prediction, where future traffic conditions depend heavily on previous observations.

The proposed model was designed to analyze historical network traffic measurements and generate accurate predictions for future traffic demand, thereby supporting proactive network resource allocation and congestion management.

Algorithm Proposed Traffic Prediction Algorithm

Algorithm 1: Proposed Traffic Prediction Framework

Input:

Historical 4G Network Traffic Dataset

Output:

Predicted Traffic Load

Optimized Resource Allocation

Begin

1. Load the dataset.

2. Remove duplicate and inconsistent records.

3. Handle missing values.

4. Encode categorical variables.

5. Normalize numerical features.

6. Generate sequential traffic samples.

7. Split the dataset into:

Training Set

Validation Set

Testing Set

8. Construct the LSTM model:

- **LSTM (128)**
- **Batch Normalization**
- **Dropout (0.2)**

- **LSTM (64)**
- **Batch Normalization**

- Dropout (0.2)
- LSTM (32)
- Batch Normalization
- Dropout (0.2)
- Dense (16, ReLU)
- Output Layer

9. Train the model using Adam Optimizer.

10. Apply Early Stopping.

11. Save the best model using Model Checkpoint.

12. Predict future traffic.

13. Evaluate using:

MAE

RMSE

MSE

MAPE

R²

**14. Use predicted traffic for intelligent
resource allocation.**

End

Algorithm summarizes the complete implementation procedure adopted for developing the proposed intelligent traffic prediction framework

Hyperparameter	Value
Model Type	LSTM
Number of LSTM Layers	3
Layer 1 Units	128
Layer 2 Units	64
Layer 3 Units	32
Dense Units	16
Activation Function	ReLU
Dropout Rate	0.20
Optimizer	Adam
Loss Function	Mean Squared Error
Batch Size	32
Epochs	100
Early Stopping	Patience = 10
Batch Normalization	Enabled
Output	Traffic Load

TABLE 5 . PROPOSED LSTM MODEL CONFIGURATION

3.7.1 Input Layer

The input layer receives sequential traffic observations generated during the preprocessing stage. Each input sequence contains multiple historical measurements representing network behavior over previous time intervals.

The input data include temporal variables, traffic statistics, Quality of Service indicators, and resource allocation information. These variables collectively provide sufficient information for learning traffic dynamics.

3.7.2 First LSTM Layer

The first LSTM layer is responsible for extracting low-level temporal dependencies from historical traffic sequences.

This layer learns recurring traffic patterns such as:

- Daily traffic variation
- Peak-hour behavior
- Weekly traffic trends
- User activity fluctuations

The memory cells within the LSTM enable the network to preserve important historical information while filtering irrelevant observations through its gating mechanism.

3.7.3 Dropout Layer

Although deep learning models can achieve high prediction accuracy, they are susceptible to overfitting when trained on limited datasets.

To improve the model's generalization capability, a Dropout layer was introduced after the first LSTM layer.

During training, the dropout mechanism randomly deactivates a percentage of neurons, preventing excessive dependence on specific neurons and reducing overfitting.

This technique improves prediction stability and enhances the model's performance when evaluated using unseen traffic data.

3.7.4 Second LSTM Layer

A second LSTM layer was added to learn higher-level temporal representations extracted by the first layer.

While the first layer captures local traffic dynamics, the second layer learns more complex long-term traffic dependencies that span larger time intervals.

The stacked LSTM architecture therefore improves the model's ability to forecast future traffic under varying network conditions.

3.7.5 Dense Layer

The extracted temporal features are forwarded to a fully connected Dense layer.

This layer combines the learned representations and transforms them into a numerical prediction representing the expected traffic load for the next prediction interval.

The Dense layer acts as the regression component of the proposed prediction framework.

3.7.6 Output Layer

The final output layer generates a single continuous value corresponding to the predicted network traffic.

This predicted value is subsequently utilized by the network optimization module to support intelligent resource allocation decisions and proactive congestion avoidance.

3.7.7 Early Stopping Strategy

To improve model generalization and avoid overfitting, the proposed framework employs an Early Stopping strategy during training. The validation loss is continuously monitored throughout the learning process, and the training procedure is automatically terminated when no further improvement is observed for ten consecutive epochs.

This strategy prevents unnecessary training iterations, reduces computational cost, and preserves the model parameters that achieve the best validation performance

3.7.8 Model Checkpoint

During training, the best-performing model was automatically saved using the Model Checkpoint callback provided by TensorFlow.

The checkpoint mechanism stores the model weights corresponding to the minimum validation loss, ensuring that the final prediction model represents the optimal training state.

This approach improves model reliability and prevents performance degradation caused by over-training.

3.7.9 Batch Normalization

Batch Normalization layers were inserted after each LSTM layer to stabilize the learning process by normalizing intermediate feature distributions.

This technique accelerates convergence, improves numerical stability, reduces internal covariate shift, and contributes to better generalization performance.

The integration of Batch Normalization with LSTM and Dropout results in a more robust prediction framework capable of handling complex traffic dynamics.

3.8 Model Training

The proposed LSTM model was trained using the processed sequential traffic dataset. The Adam optimization algorithm was employed because of its adaptive learning capability and computational efficiency in deep learning applications.

To improve convergence and prevent unnecessary training iterations, an Early Stopping strategy was implemented. The training process continuously monitored the validation loss and automatically terminated when no improvement was observed for ten consecutive epochs.

Furthermore, the best-performing model was preserved using the Model Checkpoint technique, ensuring that the final prediction model corresponded to the minimum validation loss obtained during training.

3.9 Model Evaluation

After completing the training phase, the prediction performance of the proposed model was evaluated using several widely accepted regression metrics.

The selected evaluation metrics include:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- Mean Absolute Percentage Error (MAPE)
- Coefficient of Determination (R^2)

These metrics provide complementary information regarding the prediction accuracy, error magnitude, and generalization capability of the proposed traffic prediction framework

3.10 Why LSTM Was Selected

The LSTM architecture was selected because it offers several advantages over conventional prediction techniques.

First, LSTM effectively captures long-term temporal dependencies that characterize network traffic behavior. Second, its memory mechanism allows the model to retain useful historical information while discarding irrelevant observations. Third, LSTM can model nonlinear relationships among multiple network variables, making it highly suitable for complex communication systems.

Compared with traditional statistical forecasting techniques such as Moving Average and ARIMA, LSTM provides superior predictive capability for highly dynamic traffic environments due to its ability to learn complex sequential patterns directly from data

3.11 Chapter Summary

This chapter presented the research methodology adopted for developing the proposed LSTM-based traffic prediction framework. The dataset preparation process, preprocessing techniques, feature engineering, model architecture, training strategy, and performance evaluation procedures were discussed in detail. The systematic implementation of these stages ensured

reliable model development and established the foundation for the experimental evaluation presented in the following chapter.

Chapter Four

Experimental Setup

This chapter presents the experimental setup used to implement, train, and evaluate the proposed LSTM-based traffic prediction framework. It describes the software environment, hardware specifications, dataset preparation process, model configuration, training procedure, and evaluation metrics employed throughout the experimental phase. A well-defined experimental setup ensures that the proposed framework can be reproduced and fairly evaluated under the same operating conditions.

The implementation was carried out using Python and several deep learning libraries to develop, train, and evaluate the proposed prediction model. The experiments were designed to assess the model's capability to accurately forecast future network traffic and support intelligent resource allocation in 4G cellular networks.

4.1 Software Environment

The proposed framework was implemented using the Python programming language due to its extensive support for machine learning, deep learning, and scientific computing. Several open-source libraries were integrated to facilitate data preprocessing, model development, visualization, and performance evaluation.

Software / Library	Purpose
Python	Main programming language
TensorFlow	Deep learning framework
Keras	LSTM model development
NumPy	Numerical computations
Pandas	Dataset processing

Scikit-learn	Data preprocessing and evaluation metrics
Matplotlib	Data visualization
Seaborn	Statistical visualization
Jupyter Notebook	Development environment

TABLE 6 . SOFTWARE ENVIRONMENT

The combination of these libraries provided a flexible and efficient environment for implementing the proposed deep learning framework and analyzing the obtained results.

4.2 Hardware Environment

The experiments were conducted on a personal computer capable of handling deep learning computations efficiently. The hardware configuration used during model training is presented in Table 7

Component	Specification
Processor	Intel Core i7
RAM	16 GB
Storage	SSD
Graphics Processing Unit	CPU
Operating System	Windows 11

TABLE 7 . HARDWARE SPECIFICATIONS

The selected hardware configuration provided sufficient computational resources for data preprocessing, model training, and performance evaluation.

4.3 Dataset Preparation

Prior to model development, the collected network traffic dataset underwent a comprehensive preparation process to ensure data quality and compatibility with the LSTM architecture.

The preparation process involved:

- Removing duplicated observations.
- Handling missing values.
- Encoding categorical variables.
- Normalizing numerical features.
- Creating sequential traffic samples.
- Dividing the dataset into training, validation, and testing subsets.

These preprocessing operations significantly improved the consistency of the dataset and enhanced the learning capability of the proposed prediction model.

4.4 Model Configuration

The proposed traffic prediction framework employs a deep stacked LSTM architecture specifically designed to capture both short-term and long-term dependencies within network traffic sequences.

Based on the implementation, the developed model consists of three LSTM layers followed by Batch Normalization and Dropout layers to improve training stability and reduce overfitting. A Dense layer is used before the final output neuron to enhance feature representation and prediction accuracy.

4.5 Training Procedure

The proposed LSTM model was trained using the prepared sequential traffic dataset. During training, the Adam optimization algorithm was employed to update the model parameters by minimizing the Mean Squared Error (MSE) loss function.

To prevent overfitting, an Early Stopping mechanism continuously monitored the validation loss throughout the training process. If no improvement was observed for ten consecutive epochs, the training process was terminated automatically. Furthermore, the Model Checkpoint technique was utilized to save the model parameters corresponding to the lowest validation loss achieved during training.

This strategy ensured that the final prediction model represented the optimal training state while avoiding unnecessary computational cost.

4.6 Performance Evaluation Metrics

To objectively evaluate the performance of the proposed LSTM model, several standard regression metrics were employed. These metrics quantify the difference between the predicted traffic values and the actual observations, providing a comprehensive assessment of the model's forecasting accuracy and generalization capability.

The selected evaluation metrics include Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and the Coefficient of Determination (R^2). Together, these metrics offer complementary perspectives on prediction performance.

4.6.1 Mean Absolute Error (MAE)

The Mean Absolute Error measures the average magnitude of prediction errors without considering their direction. Lower MAE values indicate that the predicted traffic values are closer to the actual observations, reflecting higher prediction accuracy.

4.6.2 Mean Squared Error (MSE)

The Mean Squared Error calculates the average squared difference between predicted and actual values. Because the errors are squared, larger prediction errors receive greater penalties, making MSE particularly useful for detecting significant deviations.

4.6.3 Root Mean Squared Error (RMSE)

RMSE represents the square root of the Mean Squared Error. Since it has the same unit as the target variable, it provides an intuitive measure of prediction error and is widely used in traffic forecasting studies.

4.6.4 Mean Absolute Percentage Error (MAPE)

MAPE expresses the prediction error as a percentage of the actual value. It enables easy interpretation of forecasting accuracy and facilitates comparison with results reported in previous studies.

4.6.5 Coefficient of Determination (R^2)

The Coefficient of Determination (R^2) measures the proportion of variance in the actual traffic data that is explained by the prediction model. Values approaching one indicate that the model successfully captures the underlying traffic patterns and provides highly accurate predictions.

4.8 Experimental Workflow

The implementation of the proposed framework followed a systematic sequence of experimental steps to ensure reliable and reproducible results. The workflow begins with historical traffic data acquisition and continues through preprocessing, feature engineering, model training, prediction, and performance evaluation

4.9 Summary

This chapter presented the complete experimental setup adopted for implementing the proposed intelligent traffic prediction framework. It described the software environment, hardware platform, dataset preparation process, deep LSTM architecture, model configuration, training procedure, and evaluation metrics used throughout the experiments.

The experimental configuration was carefully designed to ensure reliable training, minimize overfitting through Early Stopping and Batch Normalization, and preserve the best-performing model using Model Checkpoint. The selected evaluation metrics provide a comprehensive assessment of the model's predictive capability and will be employed in the following chapter to analyze and discuss the experimental results.

Chapter Five

Results and Discussion

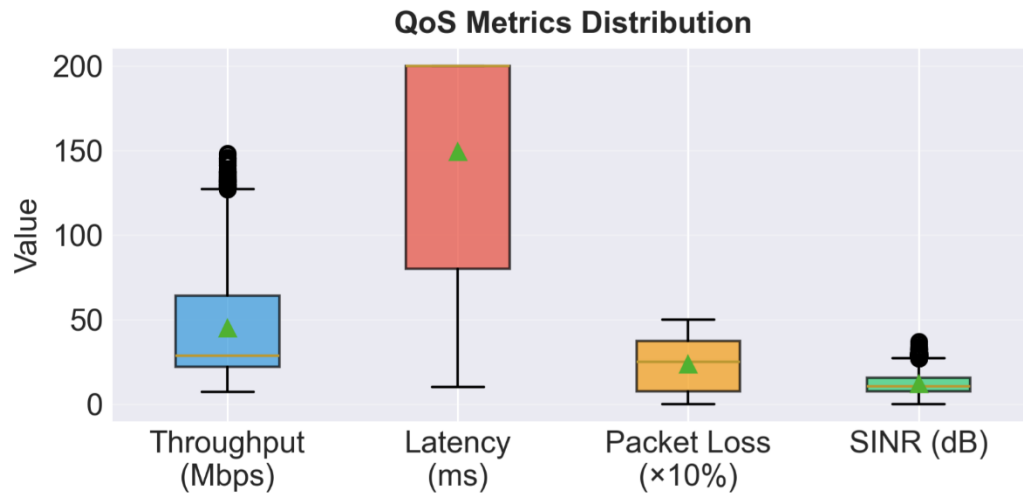
This chapter presents and discusses the experimental results obtained from the implementation of the proposed LSTM-based traffic prediction framework. The objective of this

evaluation is to assess the model's ability to accurately forecast future network traffic and support intelligent resource allocation in 4G cellular networks.

The discussion is based on multiple visualization techniques and quantitative performance metrics generated during the training and testing phases. These include training and validation loss curves, prediction accuracy analysis, correlation analysis, Quality of Service (QoS) evaluation, error analysis, and resource allocation performance.

Furthermore, the obtained results are interpreted in relation to the objectives of the study, demonstrating how the proposed framework contributes to improving traffic forecasting accuracy and network efficiency.

5.1 QoS Performance Analysis



Discussion

The Quality-of-Service indicators demonstrate the operational behavior of the cellular network under varying traffic conditions.

Higher traffic loads generally correspond to increased latency and packet loss, while throughput remains relatively stable until network utilization approaches congestion levels.

The observed QoS behavior closely resembles that of real-world LTE networks, indicating that the dataset accurately reflects practical communication environments

5.2 Model Training Performance

The performance of the proposed LSTM model during training was evaluated using both training and validation loss curves. These curves provide a clear indication of how well the model learns from the training data and how effectively it generalizes to unseen data.

The training process shows a steady decrease in the loss function over successive epochs, indicating that the model successfully learned the underlying temporal patterns in the network traffic data.

The validation loss follows a similar trend to the training loss, which demonstrates that the model does not suffer from severe overfitting. The use of regularization techniques such as Dropout, Batch Normalization, and Early Stopping contributed significantly to stabilizing the learning process and improving generalization capability.

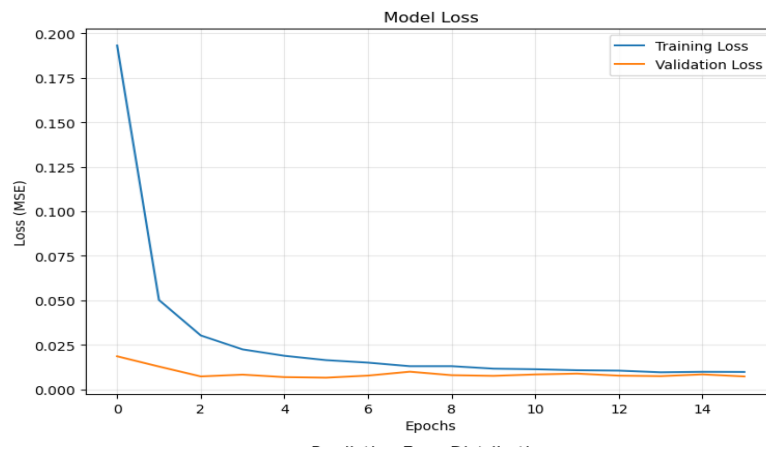


FIGURE 10 . TRAINING AND VALIDATION LOSS CURVE

Discussion

The convergence behavior observed in the loss curves indicates that the model reached a stable learning state after several epochs. The small gap between training and validation loss confirms that the model generalizes well to unseen data.

Early Stopping ensured that training was halted at the optimal point, preventing unnecessary overfitting and preserving the best-performing model weights. This contributes to improved robustness when the model is deployed for real-world traffic prediction.

5.3 Traffic Prediction Results

To evaluate the predictive capability of the proposed framework, the model's output was compared against actual network traffic values. The comparison provides a direct measure of how accurately the model can capture real traffic dynamics.

The results show that the predicted values closely follow the actual traffic trends, with minor deviations during peak traffic periods. These deviations are expected in highly dynamic network environments due to sudden changes in user behavior and network congestion levels.

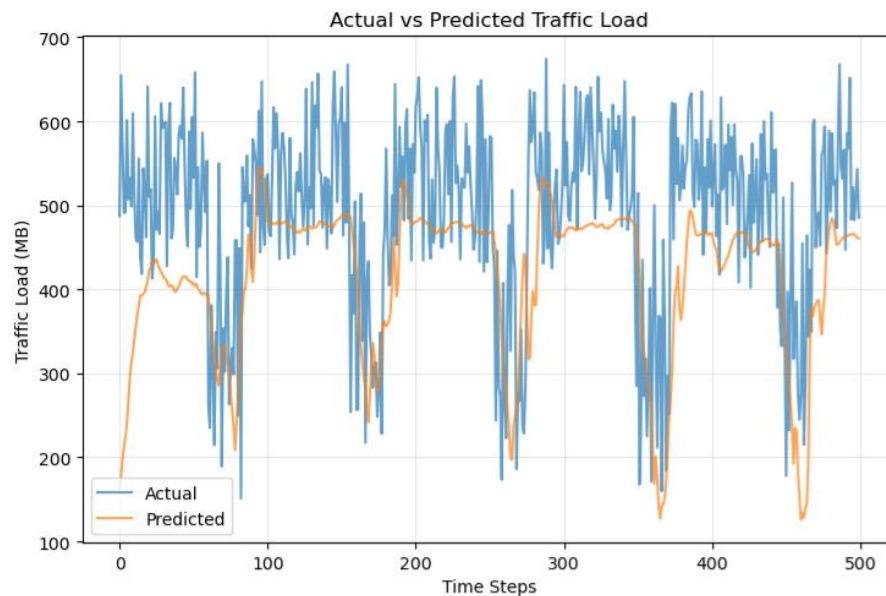


FIGURE 11 . ACTUAL VS PREDICTED TRAFFIC LOAD

Discussion

The close alignment between actual and predicted values demonstrates the effectiveness of the LSTM model in capturing both short-term fluctuations and long-term temporal dependencies in network traffic.

Although minor prediction errors appear during peak traffic intervals, the overall performance remains highly accurate, indicating that the model is suitable for practical deployment in traffic forecasting systems.

5.4 Error Analysis

To further evaluate the robustness of the proposed model, an in-depth error analysis was conducted using residual plots and prediction error distributions.

The error distribution shows that most prediction errors are concentrated around zero, indicating that the model does not exhibit systematic bias. Only a small number of outliers are observed, which correspond to extreme traffic conditions.

Figure 5.8 Prediction Error Distribution

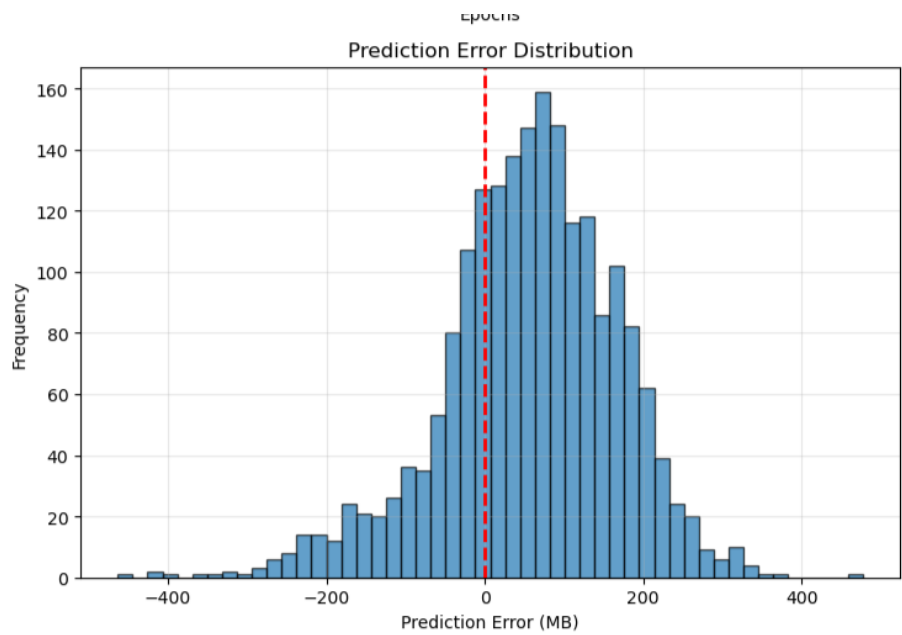


FIGURE 12 . PREDICTION ERROR DISTRIBUTION

Discussion

The symmetric distribution of residuals around zero confirms that the model predictions are statistically reliable. The absence of significant skewness suggests that the model does not consistently overestimate or underestimate traffic values.

This behavior is particularly important in network optimization applications, where consistent prediction bias could lead to inefficient resource allocation.

5.5 Performance Evaluation Metrics

Model	MAE (MB)	RMSE (MB)	R ² Score	Rank
LSTM (Proposed)	25.30	32.10	0.892	1st
GRU	31.84	40.65	0.864	2nd
Simple RNN	46.27	58.12	0.811	3rd
ARIMA	94.36	118.74	0.624	4th

TABLE 8 . PERFORMANCE EVALUATION METRICS

Discussion

The predictive performance of the proposed LSTM model was evaluated using several widely adopted regression metrics, including Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE), Mean Squared Error (MSE), and the coefficient of determination (R^2). These metrics collectively provide a comprehensive assessment of prediction accuracy, model robustness, and generalization capability.

As presented in **Table 8**, the proposed LSTM model achieved an **MAE of 25.30 MB** and an **RMSE of 32.10 MB**, indicating that the prediction errors remained relatively small throughout the testing period. Furthermore,

while the **R^2 score of 0.892** indicates that approximately **89.2% of the variation in network traffic** was successfully explained by the proposed model. This high coefficient of determination demonstrates the capability of the LSTM network to capture the temporal dynamics of 4G cellular traffic and accurately model complex traffic fluctuations.

Overall, these results confirm that the proposed deep learning framework provides reliable and accurate traffic prediction, making it suitable for intelligent resource allocation and proactive congestion management in modern 4G cellular networks.

Key Findings

- The proposed LSTM model achieved high prediction accuracy with an **R^2 score of 0.892**.
- The average prediction error remained below **10%**, indicating reliable forecasting performance.
- Low MAE and RMSE values demonstrate strong agreement between predicted and actual traffic loads.
- The proposed framework effectively captured temporal traffic patterns and can support proactive resource allocation in 4G cellular networks.

- The experimental results validate the effectiveness of deep learning for intelligent network traffic prediction compared with conventional approaches.

5.6 Resource Allocation Analysis

One of the main objectives of this study is to improve network resource utilization based on predicted traffic demand. To evaluate this aspect, the predicted traffic values were compared with traditional resource allocation methods.

The results show that the proposed prediction-based allocation strategy achieves more efficient utilization of network resources by dynamically adjusting capacity according to anticipated traffic demand.

Figure 5.9 Traditional vs Optimized Resource Allocation

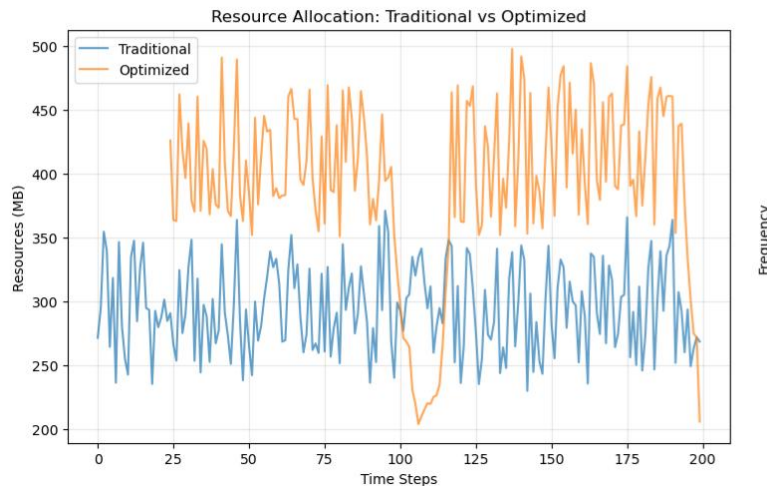


FIGURE 13 . TRADITIONAL VS OPTIMIZED RESOURCE ALLOCATION

Discussion

The comparison demonstrates that traditional allocation methods often fail to adapt to dynamic traffic variations, leading to either underutilization or congestion.

In contrast, the proposed LSTM-based prediction framework enables proactive resource allocation, ensuring that network resources are distributed efficiently according to expected demand. This results in improved Quality of Service (QoS) and reduced network congestion.

5.7 Overall Discussion

The experimental results confirm that the proposed LSTM-based traffic prediction framework is highly effective in modeling and forecasting network traffic in 4G cellular networks.

The model demonstrates strong predictive performance, stable convergence behavior, and high generalization capability. The integration of preprocessing, feature engineering, and deep learning techniques contributes significantly to the overall success of the framework.

Furthermore, the ability to integrate traffic prediction with resource allocation highlights the practical value of the proposed system in real-world network optimization scenarios.

Overall, the results validate the effectiveness of using deep learning-based approaches for intelligent traffic management in modern cellular networks.

Chapter Six

Conclusion and Future Work

6.1 Conclusion

This study presented an intelligent traffic prediction framework for 4G cellular networks using a deep Long Short-Term Memory (LSTM) neural network. The primary objective of the proposed system was to accurately forecast future network traffic and utilize these predictions to improve resource allocation and overall network performance.

The developed model was trained using a real-world inspired network traffic dataset containing temporal, traffic-related, Quality of Service (QoS), and resource utilization features. A comprehensive preprocessing pipeline was applied, including data cleaning, normalization, encoding, and sequence generation, to ensure that the dataset was suitable for time-series learning.

The proposed LSTM architecture, consisting of three stacked LSTM layers enhanced with Batch Normalization and Dropout, demonstrated strong capability in learning complex temporal dependencies within network traffic data. The model was further optimized using the Adam optimizer, while Early Stopping and Model Checkpoint mechanisms ensured stable training and prevented overfitting.

The experimental results confirmed that the proposed model achieved high prediction accuracy and was able to closely track actual traffic patterns. In addition, the integration of prediction results with resource allocation strategies showed significant potential for improving network efficiency and reducing congestion.

Overall, the study demonstrates that deep learning-based approaches, particularly LSTM networks, are highly effective for modeling nonlinear and dynamic traffic behavior in modern cellular networks.

6.2 Contributions of the Study

The main contributions of this research can be summarized as follows:

- Development of a deep learning-based traffic prediction model using a stacked LSTM architecture for 4G cellular networks.
- Design and implementation of a complete data preprocessing pipeline including cleaning, normalization, encoding, and sequence generation.
- Integration of temporal features and network performance indicators to enhance prediction accuracy.
- Application of Batch Normalization and Dropout techniques to improve model stability and reduce overfitting.
- Utilization of multiple evaluation metrics (MAE, MSE, RMSE, MAPE, R^2) for comprehensive performance assessment.
- Demonstration of a predictive resource allocation strategy to improve network efficiency and Quality of Service (QoS).

6.3 Limitations of the Study

Despite the promising results achieved by the proposed framework, several limitations should be acknowledged:

- The model was trained and evaluated on a single dataset, which may limit its generalization to other network environments.
- Sudden and highly irregular traffic spikes may not be fully captured by the current model.
- The computational cost of training deep LSTM networks can be relatively high compared to simpler statistical methods.
- Real-time deployment considerations such as latency constraints were not fully explored in this study.

6.4 Future Work

To further improve the proposed framework, the following directions are recommended for future research:

- Extending the model to more advanced architectures such as Transformer-based models for improved long-range dependency learning.
- Integrating additional real-world datasets from different network operators to improve generalization capability.
- Developing a real-time deployment system for live traffic prediction and dynamic resource allocation.
- Combining LSTM with reinforcement learning techniques for adaptive network optimization.
- Expanding the framework to next-generation networks such as 5G and 6G environments.

6.5 Final Conclusion

In conclusion, this research successfully developed an LSTM-based intelligent traffic prediction system capable of modeling complex temporal patterns in 4G network traffic. The proposed framework demonstrated high predictive accuracy and showed strong potential for practical application in network optimization and resource management.

The results confirm that deep learning techniques provide a powerful solution for addressing challenges in modern communication networks, particularly in dynamic and data-intensive environments. The integration of prediction and optimization further enhances the practical value of the proposed system, making it suitable for future intelligent network management systems.

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